8 With the rising concern in global warming, there is a renewed interest in expanding the use of renewable and sustainable energy systems. Solar-powered cars is one of the areas of innovation where researchers and car manufacturers are working on. Solar cars are electric cars that make use of photovoltaic cells or solar cells to convert energy from sunlight into electricity. These cars can store the energy in batteries so that they can run smoothly at night or in the absence of direct sunlight.

Solar powered cars can be operated using relatively low voltages that drive direct current motors and can produce reasonable power at their peak performance. The voltages are provided by a battery, which is powered by energy collected by the solar cells mounted on the roof, bonnet and boot of the car. The electrical efficiency of a typical solar cell is 26%. Electrical efficiency is the ability of the cells to convert solar energy to electrical energy. For countries near the equator, the average solar energy intensity is 0.90 kW m^{-2}. Energy intensity is the power incident per unit area of the solar cells.

The specifications for a typical solar car on an approximately 200 km daytime journey between Singapore and Malacca are as follows:

area of solar cells collecting energy	7.7 m^2
power output at peak performance	4500 W
average power output	1100 W
average speed for the journey	67 km h^{-1}
efficiency of motor at average power output	92 %
charge capacity of car battery	60 A h
operating voltage of battery	50 V

(a) Determine the total energy collected by the solar cells during its 200 km journey.

energy = ... J [3]

(b) Determine the amount of energy in **(a)** that is actually converted into electrical energy.

energy = ... J [2]

(c) (i) When the motor of the car is operating at average power output, calculate the total amount of energy consumed from the battery in one second.

energy = ... J [2]

(ii) Hence, calculate the discharge current from a fully charged battery at night.

current = ... A [2]

(d) (i) A typical solar cell has a surface area of 6.4×10^{-3} m^2.

Determine the number of solar cells fitted onto the solar car.

number of solar cells = ... [1]

(ii) Determine the amount of energy collected by each solar cell during the journey.

energy = ... J [2]

(e) (i) When the car is operating at its average power output, it travels at a constant speed of 67 km h^{-1}.

Determine the driving force produced by the engine.

driving force = ... N [2]

(ii) Determine the total resistive force at this speed.

resistive force = ... N [1]

(f) A car moving with velocity v will experience a resistive force R. Fig. 8.1 shows the variation with v^2 of R.

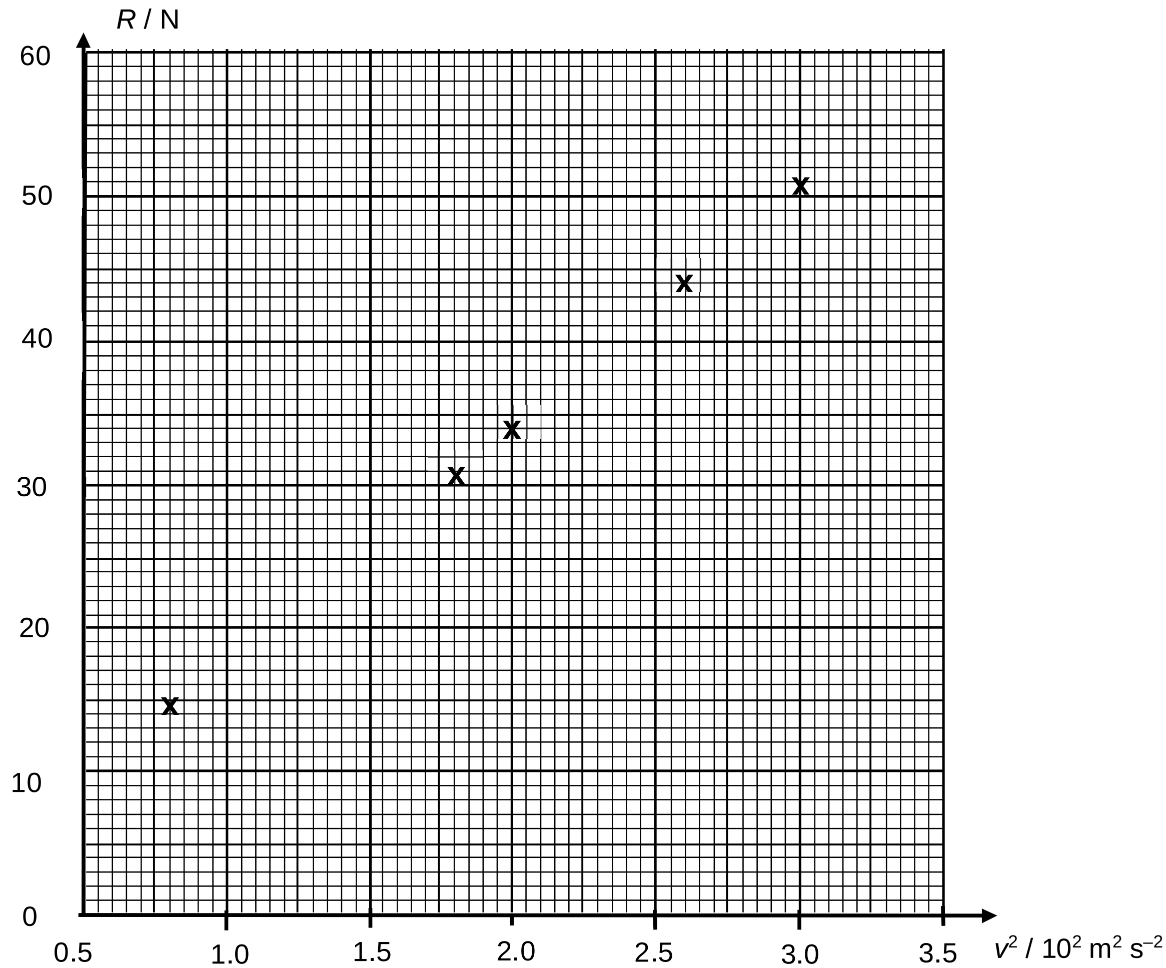

Fig. 8.1

(i) Plot the point for the car travelling at 67 km h^{-1} on Fig. 8.1. [1]

(ii) Complete Fig. 8.1 by drawing the best-fit line. [1]

(iii) State the relationship between R and v^2.

....

.... [1]

(g) The average solar energy intensity received is 0.90 kW m^{-2}.

State two factors which could affect the actual value received.

....

.... [2]

End of paper